

Effects of plant roots on soil preferential pathways and soil matrix in forest ecosystems

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Abstract To characterize effects of plant roots on preferential flow (PF), we measured root length density (RLD) and root biomass (RB) in Jiufeng National Forest Park, Beijing, China. Comparisons were made for RLD and RB between soil preferential pathways and soil matrices. RLD and RB declined with the increasing soil depth (0–10, 10–20, 20–30, 30–40, 40–50, 50–60 cm) in all experimental plots. RLD was greater in soil preferential pathways than in the surrounding soil matrix and was 69.5, 75.0 and 72.2 % for plant roots of diameter (d) <1 , $1 < d < 3$ and $3 < d < 5$ mm, respectively. Fine root systems had the most pivotal influence on soil preferential flow in this forest ecosystem. In all experimental plots, RB content was the sum of RB from soil preferential pathways and the soil matrix in each soil depth. With respect to 6 soil depth gradient (0–10, 10–20, 20–30, 30–40, 40–50, 50–60 cm) in each plot, the number of soil depth gradient that RB content was greater in soil preferential pathways than in the soil matrix was characterized, and the proportion was 68.2 % in all plots.

Keywords Preferential flow · Preferential pathways · Soil matrix · Root length density · Root biomass

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Introduction

Studies of plant roots with respect to edaphology and plant hydrological responses have been hampered by difficulties associated with mechanisms of water movement and solute transport, especially preferential flow (Bundt et al. 2001). Preferential flow describes preferential channels for water movement and solute transport from soils to roots. Pores formed by plant roots play an important role in hydrological responses. Studies of the functions of plant roots began in the early eighteenth century and gradually increased in number. Plant roots play a pivotal role in water uptake, nutrients acquisition, solute retention and soil conservation during plant growth. Plant roots grow into soil pores to form continuous channels for water and nutrient uptake (Tracy et al. 2011). The relationships between plant roots and soil preferential flow were described by Aber et al. (1985), Steudle (1994), Stokes et al. (2009), and Ceccon et al. (2011). Li and Ghodrati (1994) used breakthrough curve methods to demonstrate the effects of channels formed by plant roots on preferential transport of nitrates. Price and Hendrick (1998) found that root biomass (RB) varied by season and that living root length density (RLD) was greatest in autumn while dead RLD was greatest in winter. Volkmar (1993) confirmed that RLD had no absolute correlation with soil bulk density. Jørgensen et al. (2002) reported that soil profiles containing root channels enhanced solute transport and water movement to a greater extent than soil profiles without root channels. Dušek et al. (2006) reported that plant root zones led to more water movement. Bogner et al. (2010) used stained patterns to determine that RLD reflected preferential flow extent and that RLD was larger in preferential pathways than in the soil matrix. Both living and decayed roots provide preferred paths for soil water and solute transport (Tippkütter

1983; Angers and Caron 1998), and there is an important relationship between living and decayed roots. Bottner et al. (1999) demonstrated that living roots had effects on soil carbon metabolism during decayed root decomposition. Compared with decayed roots, living roots released more dissolved organic carbon (DOC) (Hsieh and Yang 1992) and this accelerated decomposition of dead roots. However, decayed roots prompted more water movement than living roots (Mitchell et al. 1995). Edwards et al. (1988) demonstrated that the proportion of decayed root channels of diameter <1 mm was 80 % per m^2 in soil preferential pathways.

There are increased interests in evaluating plant roots of forest ecosystems because of their role in regulating the cycling of water and nutrients for plants growth, but plant roots are difficult to measure in any forest ecosystems (Cairns et al. 1997). The upper diameter limit of fine roots varies among different studies and ranges from 1 to 5 mm. Following the original studies, data are available for two diameter limits: roots <2 mm and roots <5 mm (Kurz et al. 1996). Tufekcioglu et al. (1999) reported that fine and small roots (<5 mm) and coarse roots (>5 mm) are two major components of plant roots, and their vertical distributions modified soil physical and biological properties. Meanwhile, they sorted roots into diameter classes of 0–2 mm (fine root) and 2–5 mm (small root). Vanninen and Mäkelä (1999) classified fine root compartments as: small fine roots (diameter 0–2 mm), large fine roots (diameter 2–5 mm) and total fine roots (diameter 0–5 mm). Fine roots' share of total biomass rarely represents more than 5 % of total biomass of trees (López et al. 2001), while Santantonio et al. (1977) and Fogel (1983) reported that coarse and fine roots as a proportion of total tree biomass varied between 18 and 45 %. Brassard et al. (2011) stated that coarse root biomass (diameter >1 mm) could account for approximately 30 % of total biomass in forest ecosystems. Maybe fine plant roots has been found to vary above in relation to forest stand characteristics, i.e. tree species, stand age, density, basal area and soil properties, or environmental factors, chiefly air temperature, amount of precipitation, geographical location and elevation (Vogt et al. 1996; Jackson et al. 1997; Leuschner and Hertel 2003). Jackson et al. (1997) estimated fine root biomass and reported that live fine root biomass ranged from 130 g m^{-2} in deserts to 950 g m^{-2} in temperate grasslands.

Many studies of the relation of plant roots to preferential flow have been conducted in farmland ecosystems to characterize the effects of soil compaction, tillage systems and management on preferential flow. Few studies concentrated on forest soils that contain more plant roots and stones. Noguchi et al. (1997) reported that at least 70 % of the macropores (>2 mm) in topsoil and 55 % in subsoil in

forest soils were associated with plant roots. Hagedorn and Bundt (2002) showed that preferential flow paths in a structured forest soil persisted for decades. Beven and Germann (1982) observed that macropores formed by forest tree roots could persist for at least 50–100 years. Studies on preferential flow in forest ecosystems, especially stony lands, are few. In these systems, soil matrix flow and preferential flow are pivotal flow patterns influencing water and solute transport. In forest ecosystems, channels formed by plant roots can contribute to physical non-equilibrium at the individual plot scale (Jarvis et al. 2012). Our study aimed to determine if plant roots have greater biomass and/or occur in greater density in preferential pathways than in the soil matrix. We conducted field dye tracing experiments in a forest ecosystem in Jiufeng National Forest Park, Beijing, China, using the food dye Brilliant Blue FCF (Colour Index 42090) to trace preferential flow (stained areas) and soil matrix flow (unstained areas) (Hagedorn and Bundt 2002). Jiufeng National Forest Park is an important water conservation area which influences the groundwater security of Beijing. The objectives of our study were to: (1) compare RLD and RB of roots of diameter <1 , 1–3, and 3–5 mm between preferential pathways and the soil matrix; and (2) determine which root diameter class contributes most to preferential flow.

Materials and methods

Study area

Our study was a forest ecosystem in Jiufeng National Forest Park ($116^{\circ}28'E$, $39^{\circ}34'N$), Beijing, China. Jiufeng National Forest Park is part of Beijing Forestry University and is used for teaching and scientific research. Elevation ranges from 60 to 1,100 m a.s.l. The climate is temperate continental with mean annual precipitation of 630 mm, mean annual temperature 11.6°C , and mean annual potential evapo-transpiration of 19,000 mm. The dominant vegetation at elevations <800 m a.s.l. was plantation of *Platycladas orientalis*, *Pinus tabulaeformis*, *Quercus* spp., *Robinia pserdoacacia* containing shrubs *Prunus armniaca* and *Vitex chinensis*. Above 800 m a.s.l., *P. tabulaeformis*, *Poplar chinensis*, *Lespedeza bicolon*, *Spiraca trilobata*, *Caragana rosea* dominated the sparse forest cover. The soil has been described as sandy loam containing approximately 30 % rock fragments and gravels (Li et al. 2013).

Experimental treatment

In July 2012, we established six experimental plots within a 10×10 m quadrat situated in representative vegetation at 260 m a.s.l. Plots 1 and 2 were located in *Sophora*

japonica, plots 3 and 4 in *P. orientalis*, and plots 5 and 6 in *Quercus dentata* sections of the quadrat. Preferential flow was identified by monitoring the movement of coloured solution added to each plot. Brilliant Blue FCF dye solution (5 g L^{-1}) was applied to the experimental plots during the growth season. The solution was uniformly applied to a $1.2 \times 1.2 \text{ m}$ area centered on the experimental trees to avoid border effects (Hagedorn and Bundt 2002; Legout et al. 2009). Horizontal and vertical soil profiles were excavated when the solution had infiltrated the soil (Hu et al. 2013). Horizontal profiles were extracted from $0.5 \times 0.5 \text{ m}$ quadrats and vertical profiles with maximum dyeing depth were extracted from points centered on the experimental trees one day after dye tracer application (Hagedorn and Bundt 2002). For the horizontal and vertical sections, soil cores were extracted from preferential pathways and the soil matrix. Preferential pathways were identified by stained areas and soil matrices by unstained areas (Hagedorn and Bundt 2002). We used a camera to record preferential pathway distributions (Fig. 1).

Root parameters

RLD and RB are pivotal indices of water and solute transport in forest ecosystems, especially of preferential flow. Soil-free roots were dried for 48 h in an oven at 70°C to constant weight (Castellanos et al. 2001; Helmsaari et al. 2007) and then weighed using an electronic balance (DV215CD ($81 \text{ g}/0.01 \text{ mg}$)) to obtain plant roots. RB (g m^{-2}) (Makkonen and Helmsaari 2001) was usually measured by oven drying (Livesley et al. 1999). Fine RB is calculated on the basis of the cross-sectional area of soil

cores. RLD (total root length per soil volume) (Mosaddeghi et al. 2009; Glab 2013) was measured using WinRHIZO (STD4800) (Himmelbauer et al. 2004; Yan et al. 2011).

Root sampling

Each plot was excavated from horizontal cross sections in 10 cm depth increments 24 h after application of Brilliant Blue FCF dye solution. Undisturbed soil samples were taken at each depth using soil corers (7 cm diameter, 5 cm height, 200 cm^3 volume) with two replications in preferential pathways and the soil matrix. Samples were taken to a depth of 60 cm (0–10, 10–20, 20–30, 30–40, 40–50, 50–60 cm) in all experimental plots. Soil cores were stored at -2°C (Castellanos et al. 2001) and soil was separated from roots using 5 mm sieves. When necessary, samples were placed in dishes with 4–5 mm deep water so that roots spread and soil particles could easily be removed (Castellanos et al. 2001; Yan et al. 2011). We defined fine root diameter as $\leq 5 \text{ mm}$, as commonly used in other studies (Kurz et al. 1996; Finér et al. 2011).

Root contribution to preferential flow

The contribution of plant roots to preferential flow represented an index evaluating which kinds of root diameter functions the largest positively. Firstly, all plant roots from preferential pathways in each plot were described, and total sum of plant roots in preferential pathways obtained. Afterwards, the contribution of plant roots to preferential flow was monitored: total sum of plant roots in preferential pathways divided by plant roots $d < 1$, $1 < d < 3$, $3 < d < 5 \text{ mm}$, respectively.

Root general comparison

General comparison (GC) evaluated in this paper represented an index determining the difference of plant roots content between preferential pathways and the soil matrix. On the basis of the index, it is not complex to discriminate which one of plant roots content was greater in preferential pathways and the soil matrix. The simplified equation will be given as follows:

$$\eta = \frac{|\alpha_{PP} - \alpha_{SM}|}{\alpha_{PP}} \text{ or } \eta = \frac{|\alpha_{PP} - \alpha_{SM}|}{\alpha_{SM}} \quad (1)$$

where η is GC (%), α_{PP} is plant roots content in preferential pathways, α_{SM} is plant roots content in the soil matrix. In general, $\eta = \frac{|\alpha_{PP} - \alpha_{SM}|}{\alpha_{PP}}$ will be applied to when α_{PP} is smaller than α_{SM} , while when α_{PP} is larger than α_{SM} , $\eta = \frac{|\alpha_{PP} - \alpha_{SM}|}{\alpha_{SM}}$ will be applied to.

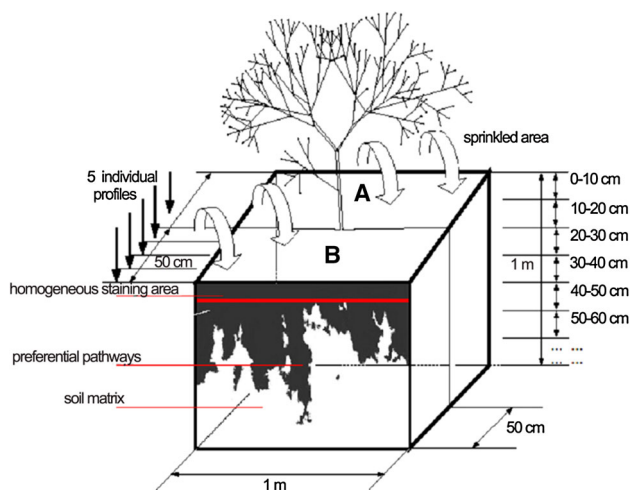


Fig. 1 Identification of preferential pathways and soil matrix from stained areas and unstained areas by applying Brilliant Blue solution. The flow patterns show the stained flow paths in black: stained areas as preferential pathways and unstained areas as soil matrix. Part A was used as a horizontal profile and part B as a vertical profile

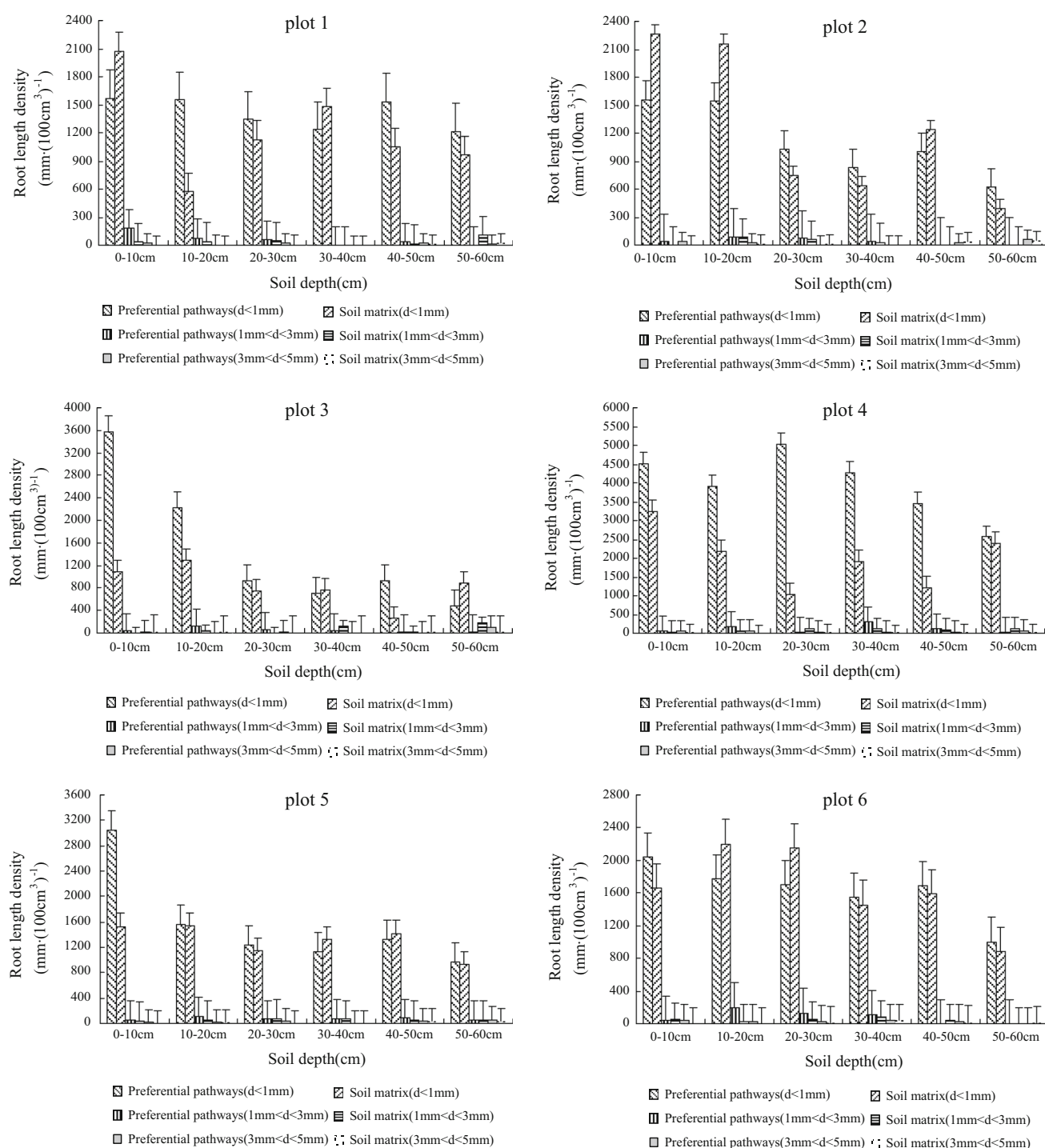


Fig. 2 Proportion of RLD by three classes of root diameter ($d < 1$, 1–3, 3–5 mm) in preferential pathways and in the soil matrix in six plots, *Sophora japonica* Linn for plots 1 and 2, *Platycladus orientalis* Franco for plots 3 and 4, *Quercus dentata* Thunb for plots 5 and 6

Statistical analysis

One-way ANOVA was used to assess differences in mean RLD and RB between preferential pathways and the soil matrix and to characterize the effects of root parameters on preferential flow. Data were analyzed using SPSS software.

Results

RLD in preferential pathways and the soil matrix

Differences in RLD by root diameter class (< 1 , 1–3 and 3–5 mm) in six experimental plots containing three types of vegetation are shown in Fig. 2. On the whole, RLD

Table 1 The proportion of the number of soil depth gradients (0–10, 10–20, 20–30, 30–40, 40–50, 50–60 cm) where RLD content was greater in soil preferential pathways than in the soil matrix among those 6 soil depth gradients (0–10, 10–20, 20–30, 30–40, 40–50, 50–60 cm) in each experimental plot

Experimental plots	Root diameter (mm)		
	$d < 1$ (%)	$1 < d < 3$ (%)	$3 < d < 5$ (%)
Plot 1	66.7	83.3	83.3
Plot 2	50.0	100	66.7
Plot 3	66.7	66.7	66.7
Plot 4	100	66.7	100
Plot 5	66.7	66.7	83.3
Plot 6	66.7	66.7	66.7

Table 2 Root contribution to preferential flow in all experimental plots

Experimental plots	Root diameter (mm)		
	$d < 1$ (%)	$1 < d < 3$ (%)	$3 < d < 5$ (%)
Plot 1	95.01	3.99	1.00
Plot 2	94.65	3.41	1.94
Plot 3	94.26	3.87	1.87
Plot 4	96.13	2.81	1.06
Plot 5	93.96	4.46	1.58
Plot 6	93.78	4.58	1.64

declined with increasing soil depth for all three root diameter classes. RLD from soil preferential pathways and the soil matrix in each soil depth also showed a similar tendency. From Fig. 2, greatest RLD was recorded in the upper soil layers to a depth 30 cm (topsoil). Meanwhile, plant roots of diameter < 1 mm were mostly distributed on the soil surface. With respect to all experimental plots, RLD content for plant roots of diameter (d) < 1 , $1 < d < 3$ and $3 < d < 5$ mm was also the sum of RLD from soil preferential pathways and the soil matrix in each soil depth. For 6 soil depth gradient (0–10, 10–20, 20–30, 30–40, 40–50, 50–60 cm) in each plot, the number of soil depth gradient that RLD content for plant roots of diameter (d) < 1 , $1 < d < 3$ and $3 < d < 5$ mm was greater in soil preferential pathways than in the soil matrix was quantified. The characterizing results were illustrated in Table 1. The proportion of RLD in preferential pathways was greater than in the soil matrix in 69.5 % of plots for roots of diameter < 1 mm, in 75 % of plots for roots of diameter 1–3 mm, and in 72.2 % of plots for roots of diameter 3–5 mm. As shown in Fig. 2 and Table 2, roots of diameter < 1 mm were the predominant component for preferential flow in all experimental plots. Roots of diameter < 1 mm accounted for almost 95.0 % of preferential flow.

RB in preferential pathways and the soil matrix

RB of fine roots ($d < 5$ mm) was densely concentrated in the upper soil layers and varied by forest type. On the whole, RB declined with increasing soil depth whether in soil preferential pathways or in the soil matrix (Table 3). In all experimental plots, RB content was the sum of RB from soil preferential pathways and the soil matrix in each soil depth. With respect to six soil depth gradient (0–10, 10–20, 20–30, 30–40, 40–50, 50–60 cm) in each plot, the number of soil depth gradient that RB content was greater in soil preferential pathways than in the soil matrix was characterized, and the proportion was 68.2 % in all plots. The difference of plant roots content (e.g., RB, $d < 5$ mm) between preferential pathways and the soil matrix was illustrated in Fig. 3. From Table 3, 22 surveyed data of RB from soil preferential pathways and the soil matrix in all experimental plots were characterized. And GC was also calculated by means of Eq. (1). Average GC was 83 % calculated from 22 surveyed data. As it was shown in Fig. 3, shaded circles represented that root general comparison of RB was higher than 83 %; and the other open circles are below 83 %. From Fig. 3, the number of shaded circles whose root general comparison of RB higher than 83 % was thirteen, while the number of open circles was nine. Those surveyed data whose GC was above 83 % accounted for 59.1 % among 22 surveyed data. However, it was ambivalent when GC was below 83 %. From Fig. 3, we implied that there were nine surveyed data whose GC was below 83 %.

Discussion

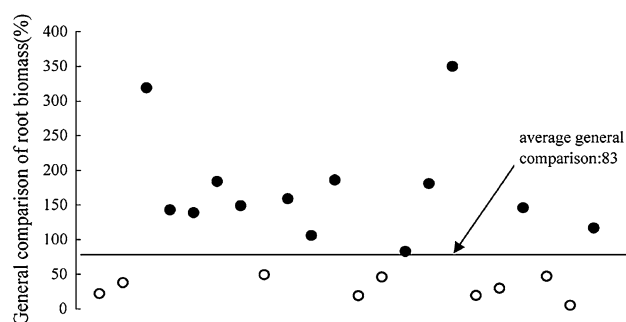
In our results, on the whole, plant roots (e.g., root length density and root biomass) declined with increasing soil depth whether in preferential pathways and in the soil matrix. These results were in agreement with Himmelbauer et al. (2010) and Bengough (2012). Meanwhile, greatest plant roots were recorded in the upper soil layer to a depth of 30 cm (topsoil). This result was similar to that reported by Lipiec et al. (2003) and Bongers et al. (2008, 2010).

Our results also confirmed that plant roots in preferential pathways were higher than in the soil matrix to some extent, particularly those distributed in the upper soil layers, because plant roots there were mostly decayed or decaying to form more preferential channels. During root decomposition, more channels are formed along the root surface. Preferential pathways provide pores or cracks and this encourages root growth. The surrounding soil matrix is too compacted for root growth. Meanwhile, our results also implied that more and more fine plant roots were located on the soil surface. This result supports the findings of Raizada

Table 3 Root biomass (g m^{-2}) in preferential pathways and the soil matrix by soil depth (0–10, 10–20, 20–30, 30–40, 40–50, 50–60 cm)

Experimental plots		Root biomass (g m^{-2}) in different soil depth (cm)					
		0–10	10–20	20–30	30–40	40–50	50–60
Plot 1	Preferential pathways	0.144	0.144	–	0.112	–	0.047
	Soil matrix	0.176	0.032	–	0.045	–	0.056
Plot 2	Preferential pathways	0.269	0.240	0.077	–	–	–
	Soil matrix	0.195	0.201	0.115	–	–	–
Plot 3	Preferential pathways	0.218	0.191	0.106	0.071	0.078	0.026
	Soil matrix	0.052	0.147	0.041	0.13	0.036	0.038
Plot 4	Preferential pathways	–	0.501	0.428	0.366	0.343	–
	Soil matrix	–	0.206	0.174	0.178	0.122	–
Plot 5	Preferential pathways	0.247	0.176	0.312	–	–	–
	Soil matrix	0.103	0.259	0.109	–	–	–
Plot 6	Preferential pathways	–	0.355	0.194	–	–	–
	Soil matrix	–	0.125	0.204	–	–	–

“–” indicates no data

**Fig. 3** Root general comparison of RB in all experimental plots. Shaded circles represent that root general comparison of RB is higher than 83 %; and the other open circles is below 83 %

et al. (2013). With respect to fine plant roots in this paper, its growth cycle is shorter than coarse plant roots. Thus fine plant roots will become decaying even decayed roots gradually. During the process, plant roots will decompose more organic matters to form more root channels.

Preferential pathways in forest ecosystems include higher organic carbon content and microbial biomass compared with soil matrix (Backnäs et al. 2012). By accumulating soil organic matters and redistributing nutrients in the soil profile, preferential pathways play significant role in their surrounding environments, particularly the soil matrix (Persson 2000). Our field experiments were carried out during heavy rain, soil water flow in preferential pathways and cracks increases, and fine roots may become asphyxiated even die. Clusters of fine roots are sometimes observed along or at the end of coarse roots and correspond to zones of major organic nutrients and water uptake. Fine roots have high decay and emission rates, and clusters may soak up water during rainy season and may contribute to decayed flow paths (Ghestem et al. 2011).

Conclusions

RLD and RB declined with increasing depth of soil. Roots were concentrated in topsoil (0–30 cm). RLD and RB were greater in topsoil than in deeper soils. Roots of diameter <1 mm accounted for the greatest proportion of all roots. RLD and RB were larger in soil preferential pathways than in the soil matrix. The contribution to preferential flow of roots of diameter <1 mm was greatest.

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